

2020-08-26

**VALENCIA FIVEPOINT RECREATION CENTER**  
**VALENCIA, CA****FEASIBILITY ASSESSMENT & DESIGN FOR GROUND SOURCE HEAT PUMP SYSTEM**

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**Project description**

We are pleased to submit this proposal to estimate the financial and technical feasibility of a ground source heat pump (GSHP) system for the Valencia Fivepoint Recreation Center, Valencia, CA. If a GSHP system is determined to be feasible, GEOOptimize will develop the design for the ground heat exchanger (GHX) and provide design assistance for the mechanical equipment and system design for the project.

The proposed recreation center includes:

- A Community Building - approximately 3,700 square feet
- Restrooms – approximately 1,900 square feet
- Pool Equipment Building – approximately 1,400 square feet
- Junior Olympic Pool – approximately 3,750 square feet
- Spa – approximately 265 square feet
- Family Pool – approximately 1,700 square feet

The GSHP system is to provide space heating and cooling for the buildings, pool water heating and DHW heating for the facility.

**System design methodology**

Optimizing the size and performance of a GHX requires a good understanding of the amount of heat rejected to the ground versus heat extracted and peak heat rejection and extraction.

**Energy modeling**

Simulating the construction and scheduled use of a building using hourly energy modeling software and weather data enables the quantification of the heating and cooling loads of the building(s) and pools needed to estimate the amount of energy transferred to and from the GHX. Modeling requires detailed information about the construction and use of the buildings and pools to:

- Develop the information needed to accurately model the design and performance of the system and GHX.
- Estimate energy performance, capital, and operating cost of a GSHP system compared to a comparable conventional system to provide heating, cooling, and pool water heating.

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## GHX modeling

Modeling the performance of a GHX requires output from energy modeling simulations of the buildings and pools as described. Input is also required from a review of the geology on the site, the land area available for construction of the GHX and discussion with geologists and / or drilling contractors about the most cost-effective drilling depth on the site based on the drilling equipment available in the region.

## Scope of work – Feasibility – Phase 1

With the information available for the building types and use, the site plans for the proposed development, and the geological / geographical information; **GEOptimize** will develop a preliminary GHX design and integration of a ground source heat pump system into the existing mechanical system. The scope of work will follow the methodology outlined above, including:

- Development of hourly energy model of the buildings based on provided drawings and occupancy information.
- Development of hourly energy models for the pools and spa based on provided drawings and use information
- Development of hourly energy load profile for domestic hot water requirements for the pool facility
- Estimated energy consumption and cost of GSHP system compared with current gas / electric heating air conditioning system both with and without the integration of the swimming pool
- Preliminary conceptual design of GHX and mechanical system for the pool buildings, pools and domestic hot water system
- Review of geological conditions at the site to estimate thermal properties of the soil, most cost-effective drilling depth and land area available
- Budget cost of construction of the GHX and GSHP system in the mechanical systems
- Presentation of findings for review for the client and design team

## Deliverables – Phase 1

**GEOptimize** will prepare a report of the findings based on the work outlined in the scope of work as a PowerPoint document. An online presentation of the report will provide an opportunity for review and questions.

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**Scope of work – Phase 2**

If Fivepoint wishes to proceed with the design and implementation of the GSHP system **GEO**ptimize will prepare:

- The detailed design of the GHX for the Valencia / Fivepoint Recreation Center system for the buildings, pools and domestic hot water loads for the facility. Final stamped drawings and specifications will be produced by a mechanical engineering firm licensed in the State of California.
- Design assistance and review of final drawings and specifications for the mechanical system schematic for the building heating and cooling systems, pool, and spa heating system and DHW system for the facility. Final stamped drawings and specifications will be produced by a mechanical engineering firm licensed in the State of California.
- Design assistance and review of final drawings and specifications of the integration of the GSHP system with the pool equipment. Final stamped drawings and specifications will be produced by a mechanical engineering firm licensed in the State of California.

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## Fee Proposal

Following is the fee schedule for the proposed energy and ground heat exchanger modeling detailed above:

| Phase              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Fee             |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1                  | <ul style="list-style-type: none"> <li>Develop hourly energy models of Community and Restroom Buildings</li> <li>Develop hourly energy models of the pools and spa</li> <li>Develop hourly energy models of DHW use for the facility</li> <li>Develop GHX model required for the combined energy loads of the buildings, pools and DHW use in the facility</li> <li>Review geological data available for the site and discussion with geologists, drilling contractors familiar with the site</li> <li>Estimate thermal performance of a GHX based on preliminary and optimized energy model, geology and borehole configuration using GLD 2016 ground heat exchanger design software. Long term performance of the GHX will be included (10-20-year performance prediction)</li> <li>Estimate energy consumption and energy cost budgets for the facility in comparison with the proposed conventional system</li> <li>Develop preliminary design of proposed GHX and distribution system piping for district system</li> <li>Develop capital cost budget estimates for construction of proposed GHX modules and district system piping</li> <li>Development of written report and PowerPoint presentation of findings               <ul style="list-style-type: none"> <li>Anticipate completion time for Phase 1: 8-10 weeks</li> </ul> </li> </ul> | \$7,500         |
| 2                  | <ul style="list-style-type: none"> <li>Develop design drawings and specifications for construction of the GSHP system and GHX by providing detailed design calculations, drawing details and specifications for the GHX. Final drawings and specifications to be completed by selected mechanical engineer licensed in the State of California</li> <li>Provide design review / design assistance for overall mechanical system design and pool equipment integration. Final drawings and specifications to be completed by selected mechanical engineer licensed in the State of California</li> <li>Provide recommendations for site inspections and review site inspection photos as needed during construction of GHX               <ul style="list-style-type: none"> <li>Anticipated completion time for Phase 2: 6-8 weeks</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | \$16,000        |
| <b>Total fees:</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>\$22,500</b> |

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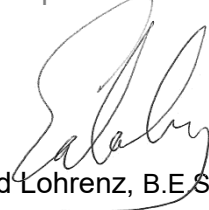
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**Closure**

Please do not hesitate to contact us if you have any questions regarding this proposal or if any clarification is required. To accept this proposal, please sign the attached signature page, and scan and email the form to me at [ed@geoptimize.ca](mailto:ed@geoptimize.ca)

Sincerely

**GE**Optimize Inc.



Ed Lohrenz, B.E.S., CGD

The scope of work and fees outlined above are acceptable to the **ENGEO Inc.** we hereby approve **GE**Optimize Inc. to proceed with the work as defined above.

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|------------------------------------------------------|----------|
| • Phase 1: Work as outlined in fee proposal          | \$6,500  |
| • Phase 2: Optional work as outlined in fee proposal | \$16,000 |

TERMS: Invoices will be submitted monthly for work completed to date. Payment is expected within 15 days of receipt of invoice.

**Project: Valencia Fivepoint Recreation Center geothermal heating and cooling system feasibility assessment and design**

Authorized Signature: \_\_\_\_\_

Name (printed): \_\_\_\_\_

Title: \_\_\_\_\_

Date accepted: \_\_\_\_\_